HumidClimatologyEngine v7.5

Detailed Scientific, Statistical, Engineering, and Reproducibility Reference

ERA5-Land hourly moisture climatology | 1981-2020 target climatology | 366-slot calendar

Release property	v7.5 contract
Primary method	Direct hourly empirical moisture climatology
Primary products	RH, vapor pressure, mixing ratio, specific humidity + moments
Primary joint product	Empirical 2-D PDF from paired hourly observations
Window model layer	Centered 5-day window for distribution fitting and station queries
Candidate univariate models	Normal, Skew-Normal, Pearson III, Beta, Bimodal Normal
Current copula evaluator	Gaussian copula (reference candidate)
Robustness	Chunked annual checkpoints + transactional bivariate progress
Progress reporting	Completed / total / percent / remaining / elapsed / ETA

Document status. This document is an implementation-aligned reference for the supplied v7.5 source and associated reporting tool. It intentionally distinguishes current implementation from future extensions.

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1. Executive Summary

Version 7.5 is a direct hourly empirical production architecture. It is intentionally different from v6: the historical v6 method operates on daily-statistical inputs and models a joint state with a Gaussian/Monte-Carlo propagation layer, whereas v7.5 starts from hourly ERA5-Land observations and computes moisture variables directly before accumulating empirical moments and joint probabilities.

The central scientific objective is to preserve the information that is lost when nonlinear moisture diagnostics are calculated only after an early daily aggregation step. The v7.5 architecture therefore treats the hourly observations as the fundamental sample population and makes distributional shape - including skewness and possible bimodality - an explicit, testable property rather than an assumption.

Core principle: the empirical bivariate product is primary. A bivariate Gaussian is only a reference evaluator; the engine does not silently assume that the joint distribution is normal.

Operational principle: long runs are partitioned into restartable spatial and temporal work units. Checkpoints are durable before progress is advanced, so a power failure should cause recomputation of at most the uncommitted work unit rather than loss of the entire climatological run.

2. Version Positioning: v6 vs v7.5

2.1 v6 historical / educational baseline

The supplied v6 source uses daily input folders and a joint state built from T, Td, and ln(P). Its covariance layer and Monte-Carlo propagation are useful as an educational baseline for dependence modeling and nonlinear uncertainty propagation. The project documentation should preserve v6 rather than delete it, while explicitly labeling it as a historical methodology rather than the preferred hourly empirical production method.

```
v6: daily-statistical inputs
-> joint state (T, Td, ln P)
-> covariance model
-> Gaussian / Monte-Carlo propagation
-> moisture diagnostics
```

2.2 v7.5 production path

The v7.5 path is fundamentally different: hourly T2m, D2m and surface pressure are transformed to RH, e, r and q at the observation level, then sufficient statistics and empirical joint probabilities are accumulated by climatological day and grid cell. A separate 5-day centered-window layer is used for local distribution fitting and station-oriented queries.

```
v7.5: hourly inputs
-> timestamp-based alignment
-> thermodynamic transformation
-> empirical moments / empirical 2-D PDF
-> 5-day centered-window model fitting (separate layer)
-> reporting / selection / diagnostics
```

2.3 What v7.5 does not claim

The source does not claim that every moisture variable is Gaussian, Beta, skew-normal, Pearson III, or bimodal. Those families are candidates for model comparison in the fitting layer. The primary joint distribution is an empirical 2-D construction, so the data can retain shapes that are not representable by a single parametric family.

3. Scientific Contract and Data Model

3.1 Target period and climatological calendar

The target climatology is 1981-2020. The project uses 366 slots. Slot 60 pools February 28 and February 29, while slot 59 remains reserved. This convention is a formal model contract.

Slot	Meaning
1-58	January 1 through February 27
59	Reserved
60	February 28 + February 29 composite
61-366	March 1 through December 31

3.2 Hourly sample definition

A primary v7.5 observation is a synchronous hourly state from temperature, dew point and surface pressure. Pairwise products require exact timestamp intersection so that the joint sample does not mix different times or marginal masks.

3.3 Validity rules

The physics layer requires finite temperature/dew point, finite positive pressure, positive vapor pressure, and $e < P$ for mixing ratio. Relative humidity is reported in the physical range while supersaturation is retained diagnostically. The distinction between reporting bounds and physical validity is preserved.

4. Input Architecture and Time Indexing

4.1 Canonical time coordinate

The 5-day extraction layer reads the actual datetime coordinate from the NetCDF dataset. The implementation recognizes valid_time and time, and otherwise searches datetime-like coordinates. File names are used for inventory, not as the source of truth for timestamps.

```
canonical_time_name(ds)
-> valid_time / time / first datetime-like coordinate
```

4.2 Grid validation

Latitude and longitude coordinates are canonicalized and checked. The production pipeline should fail closed on incompatible grids rather than silently interpolate or reshape data.

4.3 Special edge file for the start of 1981

The provided 1980-12-30 to 1980-12-31 combined file is registered as an optional edge-padding source for the first centered windows. Its observed structure is valid_time x latitude x longitude with t2m and d2m variables. This file is used only as temporal padding; it is not part of the 1981-2020 climatological target population.

4.4 End-of-period padding

The project has data through 2021-06, so the centered 5-day windows around the last days of 2020 can be completed using actual 2021 observations. Those observations provide only the required window context and do not become 2021 climatology outputs.

5. Five-Day Centered Window Engine

5.1 Definition

For target day D, the fitting population is the 5-day symmetric time window D-2 through D+2, including all hourly timestamps in the interval. The window is applied at the extraction layer, before any daily summary, so the original hourly information remains available for fitting.

```
target day D
window = [D-2 00:00, D+2 23:00]
expected hourly timestamps = 121
```

5.2 Why 121 hourly observations

A centered five-day interval contains 5 x 24 + 1 boundary-inclusive timestamps when represented as a continuous hourly interval from the beginning of D-2 to the end of D+2. The implementation records the expected count as 121 and reports completeness explicitly.

5.3 Paired T/Td alignment

Temperature and dew point are intersected by exact timestamp after extraction. This prevents accidentally combining a timestamp present in one source with a missing or shifted timestamp in the other.

5.4 Station/grid-cell access

The intended usage is a small local extraction rather than a full-grid read. The same idea can be reused to answer station queries: identify the target grid cell, load only relevant file slices, construct the centered window, and fit the requested model family.

6. Thermodynamic Transformation Layer

The v7.5 engine preserves the project phase-aware saturation formulation. Water is used for T >= 0 C and ice for T < 0 C.

Quantity	Definition / contract
Water saturation pressure	es_w(T) = 6.112 * exp(17.67*T / (T + 243.5))
Ice saturation pressure	es_i(T) = 6.112 * exp(22.46*T / (T + 272.62))
Vapor pressure	e = es(Td)
Relative humidity	RH = 100 * es(Td) / es(T)
Mixing ratio	r = 0.622 * e / (P - e)
Specific humidity	q = r / (1 + r)

RH values above 100 percent are tracked as supersaturation and clipped for the reporting field. An invalid pressure partition e >= P is not converted into a valid mixing ratio by clipping; it remains an explicit physical diagnostic.

7. Online and Mergeable Statistics

7.1 Fourth-order moment accumulation

Each DOY and spatial cell maintains count, mean, M2, M3 and M4 for moisture variables. The update is Welford/Pebay-style and therefore avoids retaining the complete hourly history in RAM.

7.2 Covariance

Selected pairwise covariance states are accumulated using mergeable updates. This supports later dependence diagnostics and comparison with copula-based representations.

7.3 Annual checkpoint semantics

Annual checkpoint files contain the sufficient state required to merge the year into the full climatology. This means a completed year can be reused without replaying its raw hourly data.

8. Primary Empirical Two-Dimensional PDF

8.1 Empirical-first principle

The primary joint probability product is an empirical 2-D piecewise-constant PDF. For the configured primary pair RH x q, hourly paired observations are binned into fixed physical edges. The density within a bin is count divided by N_valid and the bin area.

```
f(x,y) = count(x_bin, y_bin) / (N_valid * bin_area)
```

8.2 Current default pair and bins

Setting	Current default
Primary pair	RH x q
RH domain	0 to 100 percent
q domain	0 to 0.05
X bins	8
Y bins	8
Storage	hourly counts + bin edges + n_valid

8.3 Why this is not a double-normal assumption

The empirical grid retains multimodal, skewed, curved, bounded and otherwise non-Gaussian shapes. A separate bivariate Gaussian evaluator is provided only as a reference candidate and comparison tool.

8.4 Extension path

The architecture leaves room for alternative marginal distributions and dependence structures. The scientific contract should treat these as competing models rather than silently selecting a single shape for all places and days.

9. Candidate Univariate Distributions

9.1 Normal

Baseline symmetric family. It is useful as a control but should not dominate a model-selection framework simply because it is simplest.

9.2 Skew-Normal

Two fit routes are retained: a project-style moment-based parameterization inspired by the ClimateProcessingEngine convention and a likelihood-refined SciPy fit. Retaining both routes makes the fitting choice auditable.

9.3 Pearson Type III

A flexible skewed family that is especially relevant when the empirical distribution is asymmetric but not obviously bimodal. Its fit is scored alongside the other candidates using log-likelihood, AICc and BIC.

9.4 Beta

Beta is used only where the variable is bounded to [0,1], such as q when represented as a fraction or RH/100. Endpoint handling uses a small epsilon because the standard beta likelihood is defined on the open interval.

9.5 Bimodal Normal

The v7.5 fitting layer uses a 2-component Gaussian mixture. The five stored parameters are w1, mu1, sigma1, mu2 and sigma2, with w2 = 1 - w1. Component means are sorted for deterministic interpretation.

10. Bimodal Normal: Five-Parameter Model

10.1 Fit procedure

The implementation uses scikit-learn GaussianMixture with two components, multiple initializations, convergence tolerance and covariance regularization. The component order is normalized by sorting the fitted means.

Parameter	Current value
n_components	2
n_init	10
max_iter	1000
tol	1e-4
reg_covar	1e-6
random_state	20260821
Parameter count for AICc/BIC	5

10.2 Diagnostics

The fitting result retains Ashman-D-like separation and overlap coefficient in addition to the information criteria. This is important because a two-component model can reduce AIC/BIC without producing a scientifically meaningful two-regime interpretation.

10.3 Interpretation rule

BimodalNormal is a candidate, not an automatic winner. Its use is justified when fit metrics and bimodality diagnostics jointly support a genuinely two-regime shape.

11. Model Selection and Fit Scoring

The current selection rule sorts fitted candidates by minimum AICc and then BIC. The result retains the complete candidate list for auditability. This is intentionally different from hard-coding one distribution for the entire climatology.

```
data
-> fit Normal / Skew-Normal / Pearson III / Beta (where valid)
-> fit Bimodal Normal
-> compute log-likelihood, AIC, AICc, BIC
-> retain multimodality diagnostics
-> select minimum AICc, then minimum BIC
```

A future production gate can add goodness-of-fit and tail-stability criteria without changing the existing fit contract. Such an extension should be recorded as a new schema/version rather than silently altering historical selection results.

12. Copula Layer

12.1 Current implementation

The current source implements a Gaussian copula estimator. Pseudo-observations are constructed from ranks, transformed through the standard normal quantile function, and the latent correlation rho is estimated from the transformed samples.

12.2 Separation of marginals and dependence

The intended architecture is modular: fit a marginal family to each variable, then fit a dependence model to the paired samples. This avoids forcing the joint distribution to inherit the shape of a bivariate normal.

12.3 Scientific caution

A Gaussian copula is only one dependence model. Tail dependence, asymmetry and regime switching may require alternative copulas. The empirical 2-D PDF therefore remains the reference product against which copula-based approximations should be judged.

13. Restartable Processing and Power-Failure Protection

13.1 Annual checkpoint layer

The main yearly processing is written to a disk-backed NetCDF checkpoint. A year is considered complete only when its required metadata and checksum are valid.

13.2 Spatial chunking

Control	v7.5 default
CHUNK_LAT	32 grid rows
CHUNK_LON	64 grid columns
Progress flush cadence	every 16 chunks
Progress log cadence	every 8 chunks
Refresh interval	5 seconds
Maximum process workers	2

13.3 Transactional bivariate checkpoint

The empirical 2-D pass uses a per-DOY, per-spatial-chunk next_year state. The count arrays, n_valid and next_year are synchronized together. The progress marker is advanced only after the durable data update is synchronized. A power failure therefore cannot legitimately claim progress that was not committed with its data.

13.4 What gets recomputed after interruption

With chunk-level commits, an interruption should cause recomputation only of the currently uncommitted work unit. Already committed years/chunks are reused. This is a central production requirement for a multi-decade hourly job.

14. Explicit Progress, Percentage, Remaining Work, and ETA

14.1 Global progress

The run reports total work units, completed work units, percentage completed, remaining units, elapsed time and an ETA based on the observed completion rate. Runtime snapshots can also include RAM and CPU when psutil is available.

```
PROGRESS | completed=... | total=... | percent=... | remaining=... | elapsed=... | eta=...
```

14.2 Why total work units matter

A percentage without a stable denominator is misleading. The v7.5 design computes a deterministic total based on DOY and spatial chunks, so the user can see the actual fraction of the climatology completed.

14.3 Progress must never break science

Progress logging is best-effort. Failures in telemetry must not stop the numerical calculation.

15. Output Products and Naming Contract

File	Purpose
moisture_climatology_1981_2020_v7_5.nc	Primary empirical moisture climatology
moisture_climatology_diagnostics_1981_2020_v7_5.nc	Counts and physical/quality diagnostics
moisture_climatology_bivariate_1981_2020_v7_5.nc	Primary paired empirical product, configured pair
moisture_bivariate_empirical_<x>__<y>_1981_2020_v7_5.nc	Second-pass empirical 2-D PDF per pair
moisture_climatology_run_manifest_v7_5.json	Provenance, configuration and software metadata

15.1 Primary statistics

The principal scalar fields are mean, sample standard deviation, bias-corrected skewness and Fisher excess kurtosis for RH, vapor pressure, mixing ratio and specific humidity.

15.2 Bivariate fields

The empirical bivariate file stores bin edges, exact counts, valid paired observation count and the metadata needed to reconstruct the piecewise-constant joint density.

16. Bivariate Dominance Report Generator

16.1 Purpose

bivariate_dominance_report.py is a reporting-only component. It does not refit distributions or alter the model-selection results. Its job is to summarize the already-selected model code spatially and seasonally.

16.2 Report outputs

View	Question answered
Monthly share plot	Which model dominates most grid cells in each month?
DOY x model heatmap	When during the year does each model dominate?
Monthly maps	Where does each model dominate spatially?
Model legend/summary	What code corresponds to each model family?
PDF bundle	Can the result be archived as a publication/QA artifact?

16.3 Why the report matters

A daily model-selection array is difficult to inspect directly. The report generator turns it into spatial and seasonal structure, allowing researchers to identify locations and months dominated by skew-normal, Pearson III, bimodal or bounded families.

17. Station-Oriented Query Design

The 5-day window layer is intentionally reusable for a single station or grid cell. A query can be expressed as target date + latitude/longitude index, load only intersecting files, and return the centered hourly sample together with completeness metadata.

```
extract_centered_five_day_moisture_window(
target_date, lat_index, lon_index
)

returns:
hourly T2m, D2m, SP / moisture values
exact timestamps
window completeness
paired T-D count
```

The same mechanism supports interactive research: inspect a single station first, validate distributional behavior, and then run the expensive full-grid model selection only after the methodology is trusted.

18. Validation and Testing

Test	Purpose
Calendar test	Verify leap/non-leap mapping and reserved slot
Moments vs NumPy	Validate mean and central moments
Covariance vs NumPy	Validate cross-moment accumulation
Pébay merge equivalence	Confirm mergeable state matches direct accumulation
Physics test	Validate thermodynamic transformations and physical ranges
Bivariate PDF test	Verify normalization and finite output
Model-selection smoke test	Verify bimodal data can select BimodalNormal

18.1 Recommended additional QA

Before a production release, compare online states with an offline reference on small synthetic and real subsets, test restart equivalence against uninterrupted runs, audit NetCDF metadata, and verify empirical 2-D normalization numerically.

19. Reproducibility and Provenance

The run manifest records configuration hash, script SHA-256, Python and library versions, input provenance, schema version, checkpoint lineage and output locations. Reproducibility requires the exact executed configuration rather than a generic README default.

```
software version
configuration hash
script SHA-256
input inventory / identifiers
calendar convention
chunk configuration
selection settings
checkpoint lineage
final output hashes
```

20. Memory and Performance Engineering

20.1 Principle

The target is minimum wall-clock time subject to numerical correctness, memory stability and reproducibility - not maximum parallelism.

20.2 Main controls

Control	Primary effect
CHUNK_LAT / CHUNK_LON	Peak RAM and I/O granularity
MAX_WORKERS	CPU/RAM/I/O contention
Progress flush cadence	Durability vs I/O frequency
Bivariate bin count	Joint PDF resolution vs storage and counting cost

20.3 Scaling expectation

A full 1981-2020 hourly calculation over a 301 x 301 grid is dominated by file I/O, repeated spatial block access, numerical transformations and bivariate counting. Chunk sizes should therefore be benchmarked on a representative subset before committing to the full production run.

21. Scientific Interpretation and Limitations

21.1 Reanalysis limitation

ERA5-Land is a reanalysis/model product. The climatology represents the source product, not direct observational truth.

21.2 Distributional uncertainty

AICc/BIC can prefer flexible families for reasons unrelated to scientific meaning. Bimodality diagnostics and local inspection are necessary before interpreting a second mode as a physical regime.

21.3 Empirical PDF resolution

A fixed histogram is a resolution choice. Larger bin counts improve local detail but increase storage and sampling noise. Sensitivity to binning should therefore be documented.

21.4 Copula limitation

The current copula implementation is Gaussian. It is a candidate model, not a universal statement about dependence. The empirical 2-D product is the nonparametric reference.

21.5 Five-day window limitation

The centered window is a local smoothing/conditioning device for distribution fitting and station queries. It should not be interpreted as a replacement for the full climatological-day empirical population unless the scientific analysis explicitly chooses that definition.

22. Recommended Production Workflow

```
1. Inventory 1980-12 edge file and 2021 padding availability
2. Validate time axes, grids and units
3. Run mandatory unit tests
4. Run a small real-data pilot on a few grid cells
5. Inspect empirical 2-D PDF normalization
6. Fit 5-day window candidate distributions
7. Validate Bimodal / Skew-Normal / Pearson III behavior
8. Validate station extraction
9. Start full annual checkpoint accumulation
10. Run second-pass empirical bivariate PDFs
11. Run model-selection summaries
12. Generate bivariate dominance report
13. Verify restart equivalence and output hashes
14. Archive code + README + PDFs + manifests
```

Acceptance rule: a run is not publication-ready merely because it finished. The final decision should consider scientific diagnostics, restart integrity, model-selection stability and provenance completeness.

23. Release Synchronization

The intended release unit is source code + configuration + README + detailed scientific PDF + reporting tool + tests + provenance manifest + output schema. When a formula, parameter, calendar rule or output field changes, the documentation should be regenerated in the same release.

23.1 Files covered by this reference

Artifact	Role
moisture_climatology_v7_5.py	Primary implementation
README_v7_5.md	Human-readable project contract
bivariate_dominance_report.py	Visual model-dominance reporting
This PDF	Detailed scientific and engineering reference
v6 code/docs	Historical / educational baseline

24. Appendix A - Configuration Reference

Parameter	Meaning
START_YEAR / END_YEAR	Target climatology period
DOY_COUNT	366-slot climatological calendar
CHUNK_LAT	Latitude block size
CHUNK_LON	Longitude block size
MAX_WORKERS	Parallel year workers
BIVARIATE_PAIRS	Joint variable pairs
BIVARIATE_NX / NY	Empirical PDF bin counts
FIT_MIN_OBS	Minimum samples for distribution fitting
BIMODAL_N_INIT	EM initialization count
BIMODAL_MAX_ITER	EM iteration limit
BIMODAL_TOL	EM convergence tolerance
BIMODAL_REG_COVAR	Covariance regularization
WINDOW_HALF_WIDTH_DAYS	2 days on either side
WINDOW_SIZE_DAYS	5-day centered window

25. Appendix B - Data and Output Schema

25.1 Main climatology dimensions

```
doy x latitude x longitude
```

25.2 Empirical bivariate dimensions

```
doy x latitude x longitude x x_bin x y_bin
```

25.3 Progress dimensions

```
doy x y_chunk x x_chunk -> next_year
```

25.4 Key metadata concepts

Schema version, configuration hash, purpose, pair name, fixed physical ranges, PDF definition and restart definition are stored so that downstream tools can reconstruct the meaning of the file without relying on external memory.

26. Appendix C - v6 Lessons Preserved for Education

v6 is retained because it shows a transparent sequence of daily-statistical modeling, covariance construction, Gaussian propagation and nonlinear transformation. It is useful for teaching why early daily aggregation can be scientifically consequential and why nonlinear moisture variables deserve careful treatment.

The correct documentation stance is neither to hide v6 nor to present it as the production truth: preserve it as a historical baseline, document its limitations, and use v7.5 as the direct hourly empirical path.

27. Final Assessment

v7.5 is structurally much closer to a research-grade production engine: it starts from hourly observations, uses empirical moments, provides a primary empirical 2-D joint product, separates parametric fitting from the empirical reference, supports a centered 5-day fitting window, retains a five-parameter bimodal model, records detailed provenance, and uses restartable chunked computation.

The remaining scientific work is not to decide in advance which distribution is universally correct. The correct goal is a transparent competition among plausible families, evaluated per day and location, with the empirical 2-D product serving as the nonparametric reference and the dominance report showing where each model is actually useful.

Bottom line: v6 remains a valuable educational baseline; v7.5 is the production-oriented hourly empirical architecture; the bivariate dominance report turns model selection into a spatial and seasonal scientific result.

28. Appendix D - Source Function Map

Layer	Principal functions
Configuration / provenance	Config, hash_dict, sha256_file, script_sha256
Calendar	get_clim_doy, calendar_labels
Input / grid	build_file_index, open_dataset, validate_time_axis, validate_grids_and_axes
Physics	saturation_vapor_pressure, derive_moisture
Moments	update_moments_4_order, update_covariance, combine_moments, combine_covaria
2-D PDF	bivariate_histogram_edges, empirical_bivariate_histogram, empirical_bivariate_pdf
Checkpoints	year_paths, is_year_complete, create_year_checkpoint, _write_progress
Bivariate second pass	build_empirical_bivariate_pair
Five-day window	canonical_time_name, extract_centered_five_day_window, extract_centered_five_da
Model selection	_fit_standard_candidates, _fit_bimodal_normal, fit_univariate_distribution_set
Copula	pseudo_observations, fit_gaussian_copula
Validation	test_calendar, test_moments_against_numpy, test_covariance_against_numpy, test_

End of reference. The executed run configuration is authoritative; this document describes the v7.5 implementation contract represented by the supplied source at generation time.